Security Engineering in de Praktijk

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Security Engineering in de Praktijk

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$ cat passwd

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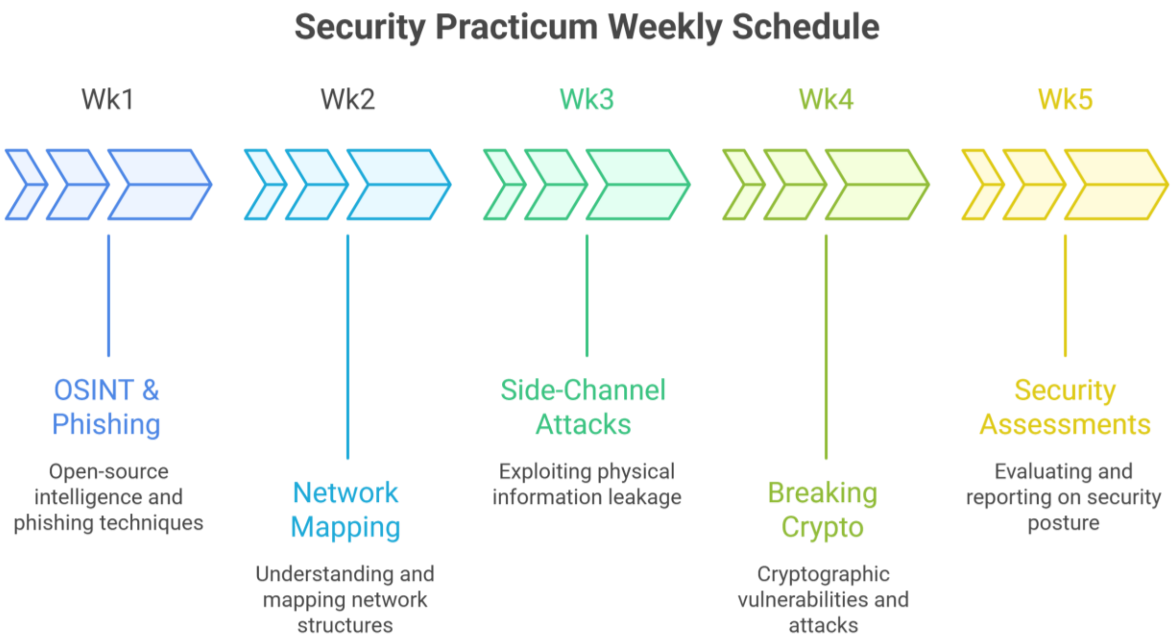
Security Practicum



Security Practicum

Wk1 – OSINT & Phishing

Wk2 – Network Mapping

Wk3 – Side-Channel Attacks

Wk4 – Breaking Crypto

Wk5 – Security Assessments

Week 2 – Network Mapping

- ❑ Network Scanners
- ❑ Building Your Own Tools
- ❑ Weekly Assignment: Network Mapping

Network Scanners

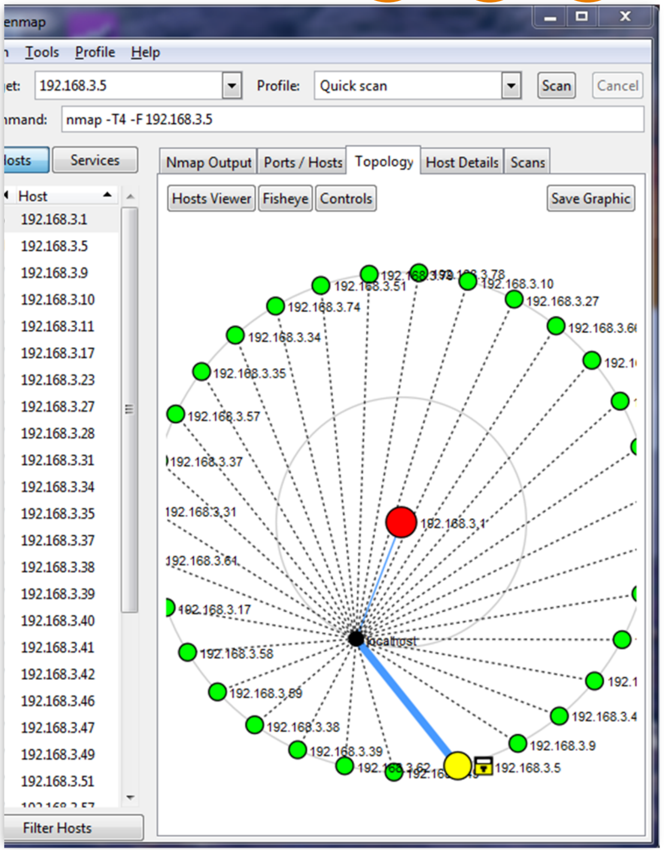
- ❑ Functionality?
  - ➢ Host Discovery
  - ➢ Port Scanning
  - ➢ Application & Version Detection
  - ➢ OS Detection
- ❑ Countermeasures
- ❑ Detection Evasion

Network Scanners

- ❑ **Localhost**
- ❑ **Green nodes**
- ❑ **Connections**
- ❑ **Red or colored nodes**
- ❑ **Network Mapping Purpose**

This visualization helps security engineers understand:

- ➢ network structure and topology
- ➢ reachable systems
- ➢ possible attack surfaces
- ➢ entry points for security testing



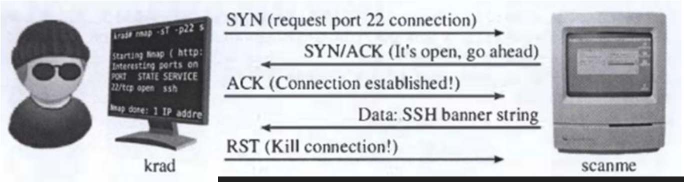
Host Discovery

- ❑ TCP SYN
- ❑ TCP ACK
- ❑ UDP Ping
- ❑ ICMP Ping
- ❑ IP Ping
- ❑ ARP Scan
- ❑ Reverse DNS

```
# nmap -sn -T4 www.lwn.net/24

Starting Nmap ( https://nmap.org )
Host 66.216.68.0 seems to be a subnet broadcast address (returned 1 extra ping)
Host 66.216.68.1 appears to be up.
Host 66.216.68.2 appears to be up.
Host 66.216.68.3 appears to be up.
Host server1.camnetsec.com (66.216.68.10) appears to be up.
Host akqa.com (66.216.68.15) appears to be up.
Host asria.org (66.216.68.18) appears to be up.
Host webcubic.net (66.216.68.19) appears to be up.
Host dizzy.yellowdog.com (66.216.68.22) appears to be up.
Host www.outdoorwire.com (66.216.68.23) appears to be up.
Host www.inspectorhosting.com (66.216.68.24) appears to be up.
Host jwebmedia.com (66.216.68.25) appears to be up.
[...]
Host rs.lwn.net (66.216.68.48) appears to be up.
Host 66.216.68.52 appears to be up.
Host cuttlefish.laughingsquid.net (66.216.68.53) appears to be up.
[...]
Nmap done: 256 IP addresses (105 hosts up) scanned in 12.69 seconds
```

Port Scanning



- ❑ **TCP**
  - ➢ SYN Scan
  - ➢ ACK Scan
  - ➢ FIN, NULL, Xmas Scans
  - ➢ Connect Scan
  - ➢ Window Scan
  - ➢ Maimon Scan
  - ➢ Idle Scan
  - ➢ FTP Bounce Scan
- ❑ **UDP Scan**
- ❑ **IP Protocol**

```
# nmap -sO 62.233.173.90 para

Starting Nmap ( https://nmap.org )
Nmap scan report for ntwklan-62-233-173-90.devs.futuro.pl (62.233.173.90)
Not shown: 240 closed ports
PROTOCOL STATE         SERVICE
1        open          icmp
4        open|filtered ip
6        open          tcp
8        open|filtered egp
9        open|filtered igp
17       filtered      udp
47       open|filtered gre
53       filtered      swipe
54       open|filtered narp
55       filtered      mobile
77       filtered      sun-nd
80       open|filtered iso-ip
88       open|filtered eigrp
89       open|filtered ospfigp
94       open|filtered ipip
103      filtered      pim

Nmap scan report for para (192.168.10.191)
Not shown: 252 closed ports
PROTOCOL STATE         SERVICE
1        open          icmp
2        open|filtered igmp
6        open          tcp
17       filtered      udp
MAC Address: 00:60:1D:38:32:90 (Lucent Technologies)

Nmap done: 2 IP addresses (2 hosts up) scanned in 458.04 seconds
```

Application and Version Detection

- ❑ Port/service lookup table
- ❑ Hostnames
- ❑ Service banners
- ❑ Service-specific probes

```
# nmap -sV -T4 -F insecure.org

Starting Nmap ( https://nmap.org )
Nmap scan report for insecure.org (74.207.254.18)
Host is up (0.016s latency).
rDNS record for 74.207.254.18: web.insecure.org
Not shown: 95 filtered ports
PORT     STATE  SERVICE  VERSION
22/tcp   open   ssh      OpenSSH 4.3 (protocol 2.0)
25/tcp   open   smtp     Postfix smtpd
80/tcp   open   http     Apache httpd 2.2.3 ((CentOS))
113/tcp  closed auth
443/tcp  open   ssl/http Apache httpd 2.2.3 ((CentOS))
Service Info: Host:  web.insecure.org

Nmap done: 1 IP address (1 host up) scanned in 14.82 seconds
```

OS Detection (Operating System Detection)

- ❑ TCP, UDP, ICMP Probes
- ❑ Fingerprint of all probes
- ❑ Small differences in RFC implementations

```
# nmap -O -v scanme.nmap.org

Starting Nmap ( https://nmap.org )
Nmap scan report for scanme.nmap.org (74.207.244.221)
Not shown: 994 closed ports
PORT      STATE    SERVICE
22/tcp    open     ssh
80/tcp    open     http
646/tcp   filtered ldp
1720/tcp  filtered H.323/Q.931
9929/tcp  open     nping-echo
31337/tcp open     Elite
Device type: general purpose
Running: Linux 2.6.X
OS CPE: cpe:/o:linux:linux_kernel:2.6.39
OS details: Linux 2.6.39
Uptime guess: 1.674 days (since Fri Sep  9 12:03:04 2011)
Network Distance: 10 hops
TCP Sequence Prediction: Difficulty=205 (Good luck!)
IP ID Sequence Generation: All zeros

Read data files from: /usr/local/bin/../share/nmap
Nmap done: 1 IP address (1 host up) scanned in 5.58 seconds
           Raw packets sent: 1063 (47.432KB) | Rcvd: 1031 (41.664KB)
```

Building Tools

- ❑ Sketching the use case
- ❑ Describing functionality
- ❑ Writing pseudocode
- ❑ Skeleton code
- ❑ Building the code

Weekly Assignment: Network Mapping

- Build your own network scanner

  - Host Discovery – Identify which devices are active in the network.
  - Port Scanning – Detect open ports and available network services.
  - Service Detection – Determine which services are running on the target systems.
  - OS Detection – Identify the operating system of the discovered devices.

Let's get started

Assignment 2
Network Mapping

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